

Men Who Made History

HENRY

FORD:

Highlights of His Life

A Publication of
The Henry Ford Museum and Greenfield Village
DEARBORN, MICHIGAN

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HIGHLIGHTS OF HIS LIFE ***

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HENRY FORD:

Highlights of His Life

A Publication of
**The Henry Ford Museum and Greenfield
Village**
DEARBORN, MICHIGAN

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The Edison Institute
Dearborn, Michigan

Henry Ford spent his early life on a farm. He was born in a small frame house that stood in a grove a few miles from Detroit, near the River Rouge. On each side of the river were the farms of people who had come to the Middle West to get land of their own. Henry's father, William Ford, was one of these early settlers. 1

The elder Ford came to America from Ireland in 1847, the year of the great potato famine in that country. He made his way to southern Michigan where he found work. At first, he labored on the railroad and then at the arsenal in Dearbornville. Later, he was a "hired hand" on the farm of Patrick O'Hern in Springwells Township. Here William Ford settled down. He purchased forty acres of land, and with his employer, Patrick O'Hern, built a farmhouse. In 1861 he married Mary Litogot, the foster daughter of O'Hern.

Henry Ford, the first son of Mary and William, was born in 1863. Soon there were brothers and sisters. Their life on the farm was a round of doing chores, working in the fields, and taking trips to town. In the winter, the children went to the one-room school over at the "Scotch settlement."

Henry Ford might well have remained on the farm and followed in his father's footsteps. The soil around Dearborn was fertile, and the products of

the land found a good market in nearby Detroit, a busy lake port. Although farming meant hard work and long hours, it was an honorable trade. William Ford was not rich, but neither was he poor. He held a position of respect in the community.

But Henry Ford had other ideas. He wanted to have something to do with machinery. He was interested in the tools of the farm rather than in the 2 farm itself. He also tinkered with watches; at the age of thirteen he was repairing the timepieces of his friends. It was a real thrill for him when Fred Reden, a neighbor, brought the first portable steam engine to Dearborn. Henry was permitted to fire its boiler. On the trips to town with his father, he saw other machines, road engines, carding mills, and grist mills.

When Henry finished school at the age of sixteen, he did what thousands of other farmers' sons were doing—he left for the city.

Detroit was a bustling town of one hundred thousand people. It was a commercial center for the Great Lakes shipping trade, and it was also an industrial center. It was to the shops and factories of Detroit that Henry Ford came to learn a trade.

His first job, at the Michigan Car Works, lasted only six days, but he soon found another one at the machine shop of James Flowers and Bros., where he became a machinist's apprentice. In this shop he learned about engines, and about the tools and machines that made parts for other machines. At night he repaired watches in the jewelry shop of James Magill. A few months later, he left the Flowers' shop to work at the Dry Dock Engine Company where he had greater opportunity to learn about steam engines. When his apprenticeship was over, he became Henry Ford, the machinist.

Instead of continuing to work at his trade in the shops of Detroit, he went back to Dearborn. William Ford lent his son eighty acres of timbered land, and Henry Ford set up a saw mill. During the harvesting season, he operated an engine for a group of threshers. He also found time to travel about southern Michigan repairing Westinghouse portable steam engines.

Henry Ford was also continuing his education. For several months he attended a business college in Detroit, all the while tinkering with 3

machines and learning still more about tools. He had an opportunity to repair, and thus become familiar with, a “gas engine” at an iron works in Detroit. As an apprentice, he had read about such an engine, invented by a German named Otto, which received its power from an explosion inside the cylinder. Several Americans were also thinking about this “internal combustion” engine.

In 1888 he married Clara Bryant, whose father owned a farm near the Ford homestead in Dearborn. After their marriage, the young couple built a “square” house nearby, where they lived for nearly three years. During this time, Henry farmed as well as repaired and operated steam engines.

In 1891, Henry Ford ended his days on the farm by moving back to Detroit, where he was employed by the Edison Illuminating Company, one of several companies that furnished electricity to the city. Detroit was changing—in the days of his apprenticeship, there had been no electric lights for homes and streets and shops. Henry’s new job was to keep the steam engines running at the Willis Avenue power station.

The year 1893 was a good year for Henry Ford, in spite of the fact that it was a hard year of panic for many other people. At the World’s Fair in Chicago that summer, he was able to add to his knowledge of gas engines and “horseless carriages.” Late in the year, his son Edsel was born. Shortly after this, his pay was raised and he was transferred to the main plant of the Illuminating Company. To be nearer his new job, the Ford family moved to 58 Bagley Avenue in Detroit. In the back yard of the new home was a brick shed; here Henry set up his tools and continued his tinkering with gas engines.

His first experiment was not long in the making. With a piece of gas pipe, an old wheel, some wire, and other scraps of metal assembled on a long board fastened to the kitchen sink in the Ford home, he made his first model 4 gas engine. Although it sputtered and jumped, it worked.

During the years that Henry Ford worked on his homemade engines, other men were also experimenting with gas engines, and they were just as determined as he was to make them operate successfully. The fair in Chicago had aroused the curiosity of many people, and the newspapers began to carry

stories about “horseless carriages.” From France came word of the exploits of Daimler and Benz with their “road wagons,” and in 1895, New Yorkers saw three Benz “horseless wagons.” Henry Ford traveled East to see them. Late in the same year, the Chicago *Times-Herald* announced that it would give a \$5,000 prize to the winner of a race between these new contraptions. Only four cars were ready at race time, and only two got away from the starting line—a Duryea and a Benz. The Benz won. One of Henry Ford’s friends, Charles B. King of Detroit, was an umpire of the race. Back in Detroit, the two friends probably discussed the merits of the cars that King had seen in Chicago. At any rate, King gave Ford some intake valves for his engines.

Henry Ford continued to add to his knowledge of gas engines. Soon after the race in Chicago, the *American Machinist* magazine told of an engine invented by E. J. Pennington which, in a day when most gas engines were bulky and heavy, was light and compact. This idea appealed to Ford, and his work took a new turn.

The First *Ford* Takes to the Road

By the spring of 1896, he was ready to make a trial run with his own horseless carriage. First he had to tear out part of the brick wall of his shed in order to get his machine into the alley. Once in the open, the engine was started, and the car bumped down the cobblestone street and continued successfully on its first short run around the block. A few weeks later, he drove it out to the homestead in Dearborn. His father was not impressed with the contraption.

These first tests meant more labor; hours of changing, adjusting, and repairing were ahead. During the day, Ford was the chief engineer of the Illuminating Company, but in his spare time, he was Ford the experimenter.

The signs of the times were encouraging. The great inventor, Thomas Edison, announced in the newspapers that the horse was doomed. Up in Lansing, R. E. Olds produced his first Oldsmobile. Alexander Winton of Cleveland drove his auto from Cleveland to New York, making what was called a “reliability” run. William K. Vanderbilt bought a car to race. All over the nation, people began to read and hear about the marvels of the new invention, of the men who were building cars, and of the races they held at fairgrounds and parks.

A new field was thus opened up for sportsmen. In Detroit, William H. Murphy, a prominent citizen, heard of Henry Ford’s car. Murphy and his friends saw an opportunity to get into the racing car business; they formed a company, called the Detroit Automobile Company, with Henry Ford as its chief engineer. At last he could leave the Illuminating Company and devote all of his time to automobiles. It was a bold move for a man with a family to give up a good job. Some of his friends and relatives told each other that he should have stayed where he was. 6

The original investment of Murphy and his friends was not enough. Improvements and changes in the model were made. Even though \$68,000

was invested in the company, the hoped for production of many cars never materialized, and in January, 1901, Henry Ford left the Detroit Automobile Company.

Henry Ford went to work on a racer which he hoped would bring him new opportunities to manufacture automobiles. His chance came in 1901. It was announced in Detroit that Alexander Winton would race his world champion car, “The Bullet,” at the Grosse Pointe race track, a few miles from the city; Henry Ford challenged the champion.

When the day of the race arrived, stores and shops closed, and a parade of sixty-eight cars moved out to Grosse Pointe. Three cars lined up for the ten-mile race, but only Winton and Ford got away. At the end of eight miles, Ford was trailing Winton, but then the “Bullet” began to sputter, and it limped to the finish line behind the racer built by Ford. The newspapers the next day reported that Henry Ford was now in the first rank of American “chauffeurs.”

In November, 1901, the Henry Ford Company was organized to manufacture automobiles, but the venture was short-lived, and four months later, Ford was working for himself again. In his small workshop, he went to work on two new racers, the “Arrow” and the “999.” With the help of a draftsman, a mechanic, and a retired bicycle champion, the new cars were made ready for racing.

Ford found a bicycle champion, Barney Oldfield, to pilot his “999” in the Manufacturers Challenge Cup Race at Grosse Pointe. This time his car led the field to a new record, finishing a mile in front of his competitors. 7

After the race, A. Y. Malcomson, a Detroit coal dealer, became interested in Henry Ford and his automobiles. The two men became partners in a new venture and Henry Ford began work on a “pilot model” for a new car. During the early months of 1903, more investors were found. By spring, a new company was organized to carry out the plans of Ford and Malcomson.

In June, 1903, the Ford Motor Company was incorporated. In addition to Malcomson, the original stockholders included James Couzens, an employee of Malcomson; John and Horace Dodge, the owners of a machine shop;

Albert Strelow, a contractor; John S. Gray, a banker; Vernon E. Fry, a real estate dealer; Charles H. Bennett, an air rifle manufacturer; C. J. Woodhall, a clerk; Horace H. Rackham and John W. Anderson, lawyers; and Henry Ford. Together, they had raised \$28,000 to start the new venture.

The new company rented a building on Mack Avenue in Detroit for \$75 a month and prepared to manufacture its automobiles. The new factory was 250 feet long by 50 feet wide. This was adequate space, since the new company did not attempt to make any of the parts for its cars. The Dodge brothers, who owned a large machine shop, made the Ford chassis, a carriage company built the body, and the wheels were purchased in Lansing. Once the parts were brought together, a dozen men assembled, adjusted, and tested the completed car—the early model A Ford. Soon this car, which sold for \$950 f.o.b. Detroit, was advertised as the “boss of the road.” Its two cylinders gave it a maximum speed of thirty miles per hour.

Although Henry Ford’s first two ventures into automobile manufacturing had not been successful, this third attempt showed great promise. At the end of the first year, the Ford Motor Company had sold over seventeen hundred automobiles.

The year 1903, which marked a turning point in the life of Henry Ford, 8 now forty years of age, was an interesting year for all Americans. The first successful transcontinental automobile trip was completed in August. A telegraphic cable was completed across the Pacific to Manila in the Philippines. In Washington, Theodore Roosevelt was clearing the way for the building of the Panama Canal, and in New Jersey, the Edison studios completed the first full-length motion picture, called “The Great Train Robbery.” Finally, in December, Orville and Wilbur Wright flew an airplane successfully at Kitty Hawk, North Carolina. During the next few years, those who marveled at progress in 1903 saw even greater changes in their way of living brought about by improved transportation and communication. The twentieth century had started off well.

The Ford Motor Company was also experimenting with changes. Models B, C, and F appeared. When the Mack Avenue plant was no longer adequate to house the activities of the company, a new building was erected, ten times larger than the first one. By 1906, the company announced models N, S, and

R, as well as a six-cylinder K. Experimentation went beyond the building of different models of automobiles; for example, a tractor was planned and constructed, but never sold.

During this time, changes were also made in the ownership of the plant. Some of the stockholders sold out, and Henry Ford became the major owner of the Ford Motor Company.

A real danger to Ford and his company during these early years was the threat of a patent suit. Back at the time of the *Times-Herald* race in 1895, George Selden, a lawyer in Rochester, New York, had patented a “self-propelled vehicle driven by an internal combustion engine.” Although he never built an automobile, all those who did were threatened with infringement suits. This fact made it necessary for the makers of automobiles to pay royalties to the patent holder. The Ford Motor Company refused. The result was a long and involved battle in the federal courts, lasting until 1911 when the case was finally settled in favor of the Ford Motor Company.



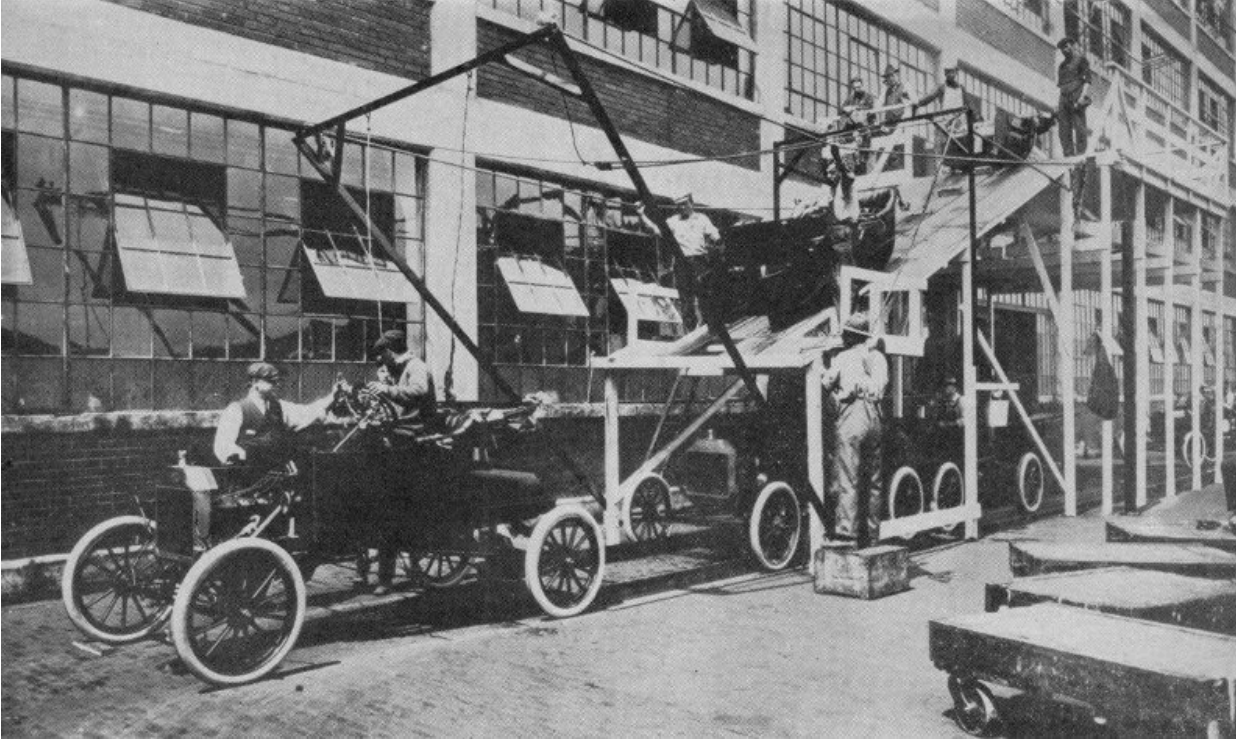
The farmhouse where Henry Ford was born in 1863 stands today in Greenfield Village ...



... along with the small brick schoolhouse from the “Scotch settlement” where he began school in 1871.

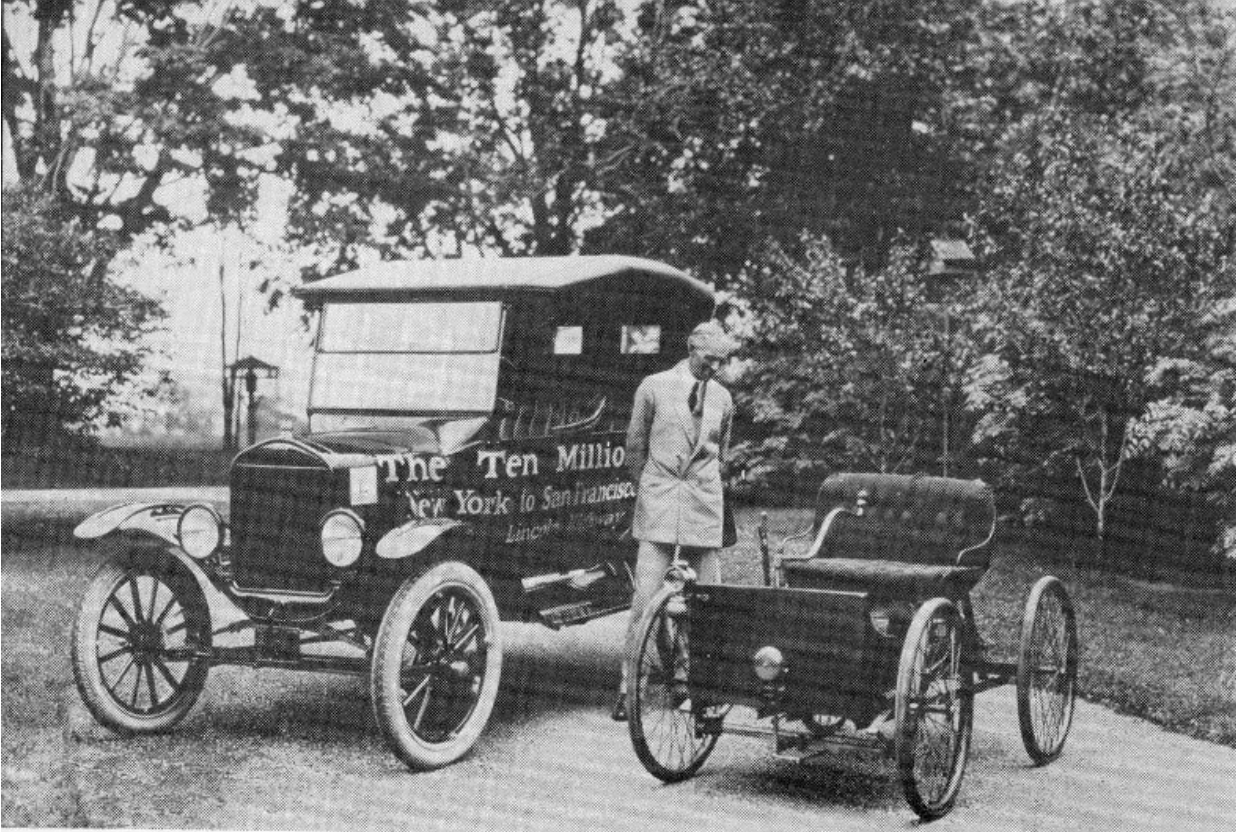


The interior of the 58 Bagley Avenue workshop where Ford built his first car, as restored in Greenfield Village. The car itself is now on display in The Henry Ford Museum.

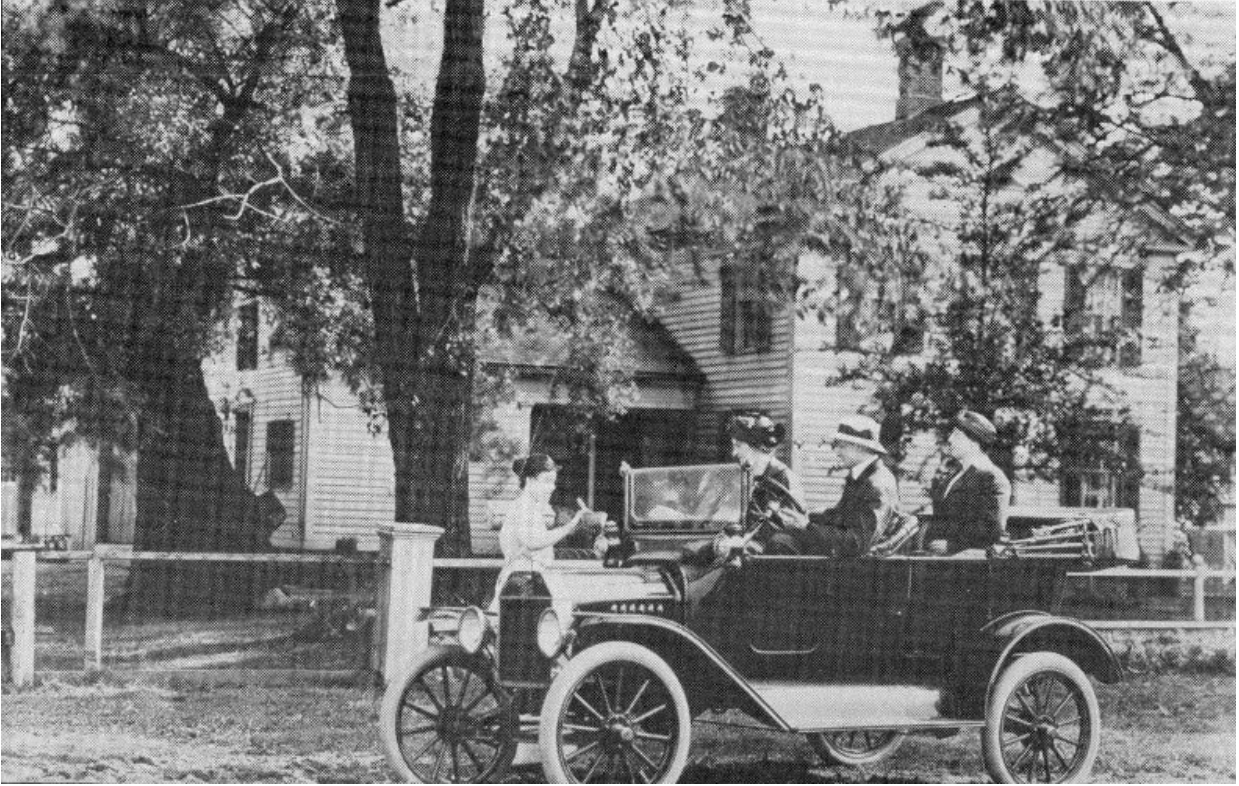


The “body drop” at the Highland Park plant of the Ford Motor Company in 1913. The modern automobile assembly line grew from such operations as this.

Photo courtesy Ford Motor Company Archives



Henry Ford views his first car and the ten millionth Ford in 1924....



By this time the Model T standing in front of a farmhouse was a familiar sight all over America.

Photo courtesy Ford Motor Company

The Model T is Born

In spite of experimentation, reorganization, and patent difficulties, the year 1907 found the Ford Motor Company operating successfully. This was a year of panic in the nation, but the company made plans to build an even bigger factory to manufacture automobiles. In 1908, it was announced that the Ford Motor Company had purchased a race track in nearby Highland Park where it would construct the largest automobile plant in the world. At the same time, draftsmen and engineers were planning a new model, to be called model T. In October, 1908, the first model T appeared, and before the year ended, over three hundred of them had been shipped to dealers throughout the nation. The Ford company claimed that this new automobile “would sound the death knell of high prices and big profits.” Its advertisements announced that “we can devote all our time and money to taking care of the orders for the car that people have actually been waiting for—a family car at an honest price.” The engine was new and “get-at-able,” an important feature in a day when most automobile owners repaired their own cars.

While Henry Ford had been a prominent automobile manufacturer before 1908, it was this new “universal” car that brought a new era in America. The automobile was no longer intended for sportsmen alone, but for all 14 those who could afford to buy one, and Henry Ford intended to see that the price of his automobile was low enough for millions of people to own a Ford car.

Soon after the success of the model T was assured, Henry Ford was asked about the secret of his ability to produce automobiles. He divulged his “secret” in these terms: have a simple design, use the latest machinery, standardize the parts, make the entire automobile yourself, and always have a good supply of materials on hand. Throughout the rest of his life, he held to these principles.

By 1911, Ford cars were manufactured by the hundreds of thousands. The process of assembling automobiles received more and more attention, and by 1914, a Ford car could be put together in an hour and a half. By the end of 1915, a million model T's had been produced. It had taken seven years to make this many of them, but in the next eleven years, fourteen million more were placed on the market. The model T Ford was the "universal car" in fact as well as in name.

While model T's changed in appearance from year to year, there was an even greater and never-ending change in the process of manufacturing them. The resulting economies brought lower prices, and this meant more cars for more people.

Henry Ford was ever on the lookout for ways and means to produce cars more efficiently. By 1914, a floor conveyor was in operation at the Highland Park plant, so that the half-completed cars moved through the plant while the workers stayed in one place. This assembly line became the key to greater production. In order to keep the line moving smoothly, machinery was continually being rearranged and new chutes and conveyors were installed. Each improvement brought new and often unforeseen problems in the never-ending task of fitting together the pieces of the huge jig-saw puzzle of production.

Although the Ford Motor Company did not manufacture the parts that 15 were used in assembling the first Ford, this policy was changed through the years. More and more of the model T was made by the company in its own plant. In 1915, Henry Ford sent an agent out to Dearborn to buy farm land along the River Rouge—thousands of acres were purchased. Now there would be room not only to enlarge the assembly line itself, but to manufacture more of the Ford in one factory. Here it would be possible to begin with raw materials—iron ore, sand, cotton, rubber, and the countless other materials—and convert them into steel, glass, and cloth to make the Ford. With this in view, a tremendous program of construction was begun along the banks of the River Rouge.

Soon there were industries within industries at the Rouge. Blast furnaces and coke ovens were fed with coal, iron ore, and limestone brought to the plant from Ford mines by Ford railroads and Ford freighters. There were glass

mills, paper factories, tire plants, and saw mills. The products of these mills and plants flowed into the assembly lines not only at the Rouge plant but at other assembly plants scattered all over the world.

The production of Ford cars rose to gigantic proportions. In 1914, the year that Henry Ford and his engineers began to plan for the plant on the Rouge, over two hundred thousand automobiles were produced. But in the year 1923, over two million model T's rolled off the assembly lines. In 1925, nearly ten thousand Fords were completed in a single day in Ford plants. The world had never before seen such an industrial giant as the one Henry Ford and his son Edsel had created.

The automobiles produced by Ford and his competitors did more than replace the horse and carriage. They changed the daily habits of Americans everywhere.

No longer did families in the cities have to live in the shadow of the factories where the head of the household was employed. Some families moved to the "suburbs" many miles away from stores and industries, while other 16 city dwellers left the urban areas completely to live in the country. On the other hand, those who had always lived on farms could come to the city with ease in automobiles. The farmers were able to enjoy the advantages of the city. Thus it was that Americans moved about in a way unheard of years before.

As travel increased, dirt roads were replaced by the super-highways. Service stations, motels, and garages dotted the newly-built concrete and brick roads. In the cities, people became aware of the "parking" problem, and of "traffic jams." The increasing number of automobiles also brought death on the highways and city streets to hundreds of Americans each year.

As the years went by, automobile manufacturers changed too. At first, there were dozens of makers of automobiles whose names are all but forgotten now. As the number of manufacturers decreased, the number of cars made annually by the remaining companies grew larger and larger. Eventually people talked about the "big three" of the automobile industry.

Competition between rival makers and Henry Ford brought the days of the model T to an end. From 1908 to 1927, fifteen million Fords had been produced, the Ford Motor Company had become the colossus of the industry, and Henry Ford's name was known all over the world. Nevertheless, in order to maintain this position, it was necessary to keep pace with the times. In order to do this, the Rouge plant was silenced until a "new" Ford could be designed and put into production. Machine tools had to be replaced, and new dies and fixtures made. To accomplish this in the largest factory in the world was a herculean task. Industrial leaders all over the nation watched with eager eyes for news from Henry Ford at Dearborn. Finally, in December, 1927, the new model A was shown to the public. This new Ford was front page news over the nation. At the Madison Square Garden in New York City, attendance records were broken when crowds came to see the model A. 17

This new Ford bore little resemblance to the model T. It now had a gear shift, four-wheel brakes, and a foot throttle. It offered many variations in body styles and color, and it was the first automobile to have a safety-glass windshield.

The River Rouge plant once again hummed with activity. Thousands of model A's were produced each day. By 1932, five million of the "new" Fords were on the highways of the nation. That year, the Ford Motor Company introduced the V-8 engine, which was unique in its field.

The transition from the gas-pipe cylinder on the kitchen sink back at 58 Bagley Avenue in the 1890's to powerful eight-cylinder engines that were turned out by the thousands each day at the largest industrial plant in the world had taken a long time to accomplish. During these decades, Henry Ford relentlessly pursued his idea—more cars for more people.

New Fields Beckon

Automobiles, however, were not the only thing that interested the man who put America on wheels. He had many other ideas, some of which failed while others succeeded.

There was, first, the matter of his interest in farming, and in the land. He had experimented with a gasoline farm tractor early in the 1900's, but he was unsuccessful in developing a line of farm engines. In 1915, a tractor 18 plant at Dearborn was begun. Some of the first tractors were sent to British farmers during the first World War. Out of this experience grew the Fordson tractor, which, like the model T, was light in weight and had a low selling price. Much more important to the farmers than his tractors was the effect that the model T Ford car had on the life of the people in rural areas. It had, in fact, been called "the farmers' car." Eventually Henry Ford became interested in farming itself. He had thousands of acres put under cultivation on Ford farms. In the 1930's, with the world burdened under an economic depression, Ford revived his interest in a program which he had begun in 1919 of establishing small factories in the rural areas of Michigan. He hoped to reverse the tide that saw farmers rushing to become city dwellers, thereby bringing about a decline of agricultural areas. He also became interested in the products of the soil that could be used in the making of automobiles. The agricultural laboratories he established discovered many substitutes for metals, and Ford cars began to appear with parts made from soybeans and other agricultural crops.

He also had an idea about transportation in the air. The idea ranged from "flivver planes" to all-metal three-motored transport ships. The Ford "trimotor" played an important part in the development of commercial aviation. Pilots called it the "tin goose," and it became known far and wide as a "safe" airplane in a day when flying was hazardous. Aviation safety was also aided when a radio beacon system was first perfected at Dearborn. Then for a period of years, Ford dropped the manufacture of airplanes, but in 1941, with the coming of World War II, airplane production became a necessary

part of war preparation, and Henry Ford made airplanes again. This time they were faster, bigger, and more deadly. Bombers moved down the assembly lines at the Willow Run plant, which was the “Rouge” of the bomber industry. When the war ended and production was halted, over 19 eight thousand “Liberators” had been assembled by the Ford Motor Company.

Henry Ford watched over the assembly line at Willow Run just as he did at the Rouge. He continued to learn from tools and machines and to “read” the story they had to tell. Out of his own experiences came his ideas about education. In speaking of his youth, he once said, “I was studying all the time, not only from books but also from things.” In 1929 he put his ideas into practice by establishing a combined school and museum, which he called The Edison Institute.

Here he brought the schoolhouses, the dwellings, the workshops, and the stores of other days so that the pupils in the schools of the Institute could study not only from books but from things. This part of the Institute he called Greenfield Village. He gathered thousands of objects to put into The Henry Ford Museum so that they could be studied by all who wanted to learn about the past. “I am collecting the history of our people as written into things their hands made and used,” he said; “... a piece of machinery or anything that is made is like a book, if you can read it. It is part of man’s spirit.” He thus provided a school not only for children but also for workers. Henry Ford once called the museum and Village that he had created “a people’s university.” Here at The Edison Institute the creator of mass production methods—now intensely interested in the hand work of village craftsmen—spent many hours. He continued to be keenly interested in the future of the automobile industry while he collected relics of the past. His unusual appreciation of both the past and the future is illustrated by the description of Henry Ford as the one man who could spend his time collecting old churns while he followed the experiments of his chemists concerning synthetic milk.

From his home in Dearborn, he kept a watchful eye over the Rouge 20 plant, now headed by his grandson, and over The Edison Institute. It was after a visit to Greenfield Village on the afternoon of April 7, 1947, that Henry Ford became ill. He died during the evening at the age of eighty-

three. This man, who had been born into a world of horse-drawn carriages, had left a world of powered wheels.

While the entire Edison Institute stands as a memorial to Henry Ford's interest in the past, certain exhibits have a particularly close relationship to his life and work.

At the Henry Ford Museum, visitors can see a special exhibition entitled "Henry Ford: a Personal History." This display depicts the important stages in Henry Ford's busy career.

In Greenfield Village, buildings which were closely associated with his life include:

The Birthplace of Henry Ford.

The Bagley Avenue workshop, where Henry Ford built his first automobile.

The Magill Jewelry Store, where, as a youth, he repaired watches and clocks.

The Scotch Settlement School and the Miller School, where Henry Ford received his formal education.

Transcriber's Notes

- Silently corrected a few typos.
- Retained publication information from the printed edition: this eBook is public-domain in the country of publication.
- In the text versions only, text in italics is delimited by _underscores_.

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